

Oracles and Computational Complexity

Comp. Complexity:

of calls to a given oracle
+
basic arithmetic operations.

The Basic Strong/Weak Problems - 1/2

Definition: Expansion and Deep Interior

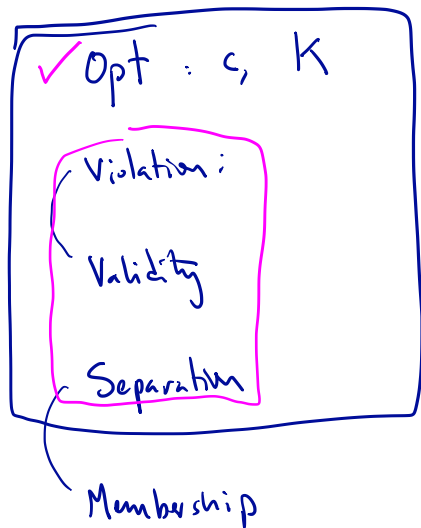
$$K \subseteq \mathbb{R}^n \quad (\text{convex})$$

x $S(x, \varepsilon)$ ball of radius ε centered at x .

$$K \subseteq S(K, \varepsilon) = \{x \in \mathbb{R}^n : \|x - y\| \leq \varepsilon, \text{ for some } y \in K\}$$

$$K \supseteq S(K, -\varepsilon) = \{x \in K : S(x, \varepsilon) \subseteq K\}$$

The Basic Strong/Weak Problems - 2/2



Optimization

SOPT:

Given $c \in \mathbb{R}^n$, find $y \in K$ that
maximizes $c^T x$ over K ,
or assert $K = \emptyset$

WOPT:

Given $c \in \mathbb{Q}^n$, and $\epsilon > 0$, $\epsilon \in \mathbb{Q}$, either

(i) Find $y \in \mathbb{Q}^n$ st
 $\forall x \in S(K, -\epsilon)$,

$y \in S(K, \epsilon)$ and $c^T x \leq c^T y + \epsilon$

or

(ii) Assert that $S(K, -\epsilon)$ empty.